

ASundar



ASD-906B User Manual

VR:1.1

Shenzhen ASUNDAR Electronics Co., Ltd

www.asundar.com

Foreword

Thank you for purchasing the products of Shenzhen ASUNDAR Electronics Co., Ltd.

This user manual describes how to use the ASD-906B analog battery product (referred to as the product in this manual). Please read this user manual carefully before use.

Cautions

- ◆ Confirm carefully when unpacking and inspecting: ①Whether the product model is consistent with your order. ②Whether the accessories such as the product certificate, user manual and warranty sheet are complete. ③Whether the product is damaged during transportation, if there is any damage, omission, damage, etc., please contact our company or your supplier as soon as possible.
- ◆ Before powering on, please pay attention to check whether the input voltage of the grid is within the range of the rated parameters. Exceeding the range may cause the product to fail to work, burn out or more serious consequences.
- ◆ make sure that the power input terminal is well grounded, otherwise it may cause electric shock.
- ◆ Due to product upgrades or specification updates, and in order to improve the accuracy of the manual, the content of this manual will be updated in time. If there is a difference with the actual use, please refer to the latest version of the manual or contact our customer service staff.
- ◆ If you still have questions after reading this manual, contact our customer service staff please.
- ◆ Customer Service Tel: 0755-89325058 4001656165

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1. Product introduction

ASD-906 B analog battery power supply has output power supply and input charging functions, which can simulate battery charging and discharging. This equipment is mainly used for the development and testing of chargers and battery-powered equipment. It can also be used as an ordinary adjustable power supply. The output voltage and input and output current protection values can be set.

Features

Can detect PCBA standby current;

With charging and discharging functions;

Fast charge and discharge switch;

High stability, low noise, low drift;

6 groups of voltage setting memory storage function ;

High precision and high resolution 1mV/1uA;

When the current is small, it can be switched to the MA gear, and the μ A level value will be displayed ;

It can visually display the curve relationship of voltage and current with time during the test;

4.3 inch touch LCD screen, support direct input of parameters ;

Support manual debugging interface, easy operation of panel functions;

It can be controlled by a PC, and can set a continuously changing dynamic voltage;

Optional serial communication version (baud rate 115200);

Wide power supply, suitable for 100~240V AC;

2. Technical specifications

Name		Analog battery	
Model		ASD906B	
Number	Types	Project	Parameter
1	Output (discharge)	Voltage	0. 1 ~20V
		Current	0~12A
2	Input (charging)	Voltage	0.2~20V
		Current	0~1 2 A
3	Set the resolution	Voltage	1mV
		Current	1mA / 1uA
4	Setting accuracy	Voltage	0.05%+2mV
5	Readback resolution	Voltage	1mV
		Current	0.01~1 2 A : 1mA ; 0~0.01A : 1uA
6	Readback accuracy	Voltage	0.05% +2mV
		Current	0.15% +4mA
7	Hello Elena	Voltage	<0.06%
8	Power regulation rate	Voltage	<0.05%
9	Ripple	Voltage	Max100mV
specification			
1	Equipment size	Power supply	220mmD*220mmW*102mmH
2	Package dimensions	Outer box	290 mmD*3 60 mmW* 180 mmH
3	Weight	Net weight	2.7Kg
		Total packaging weight	3. 4 Kg
Supplementary features			
1	Grid input	Voltage	100~240V ac
		Frequency	47~63Hz
2	Cooling method	Fan	60mm*60mm*20mm
3	Temperature	Operating environment	0 to 40 °C
		Storage environment	-20 to 70 °C
4	Use environment: indoor use design, the maximum humidity is 80% , and the instrument has no condensation.		

3.Quick start

3.1 appearance



Figure 1: Front panel

- ① Function setting area
- ② LCD display
- ③ Quick switch and start switch function area
- ④ Power switch
- ⑤ Adjusting knob (with button function)
- ⑥ Output terminal

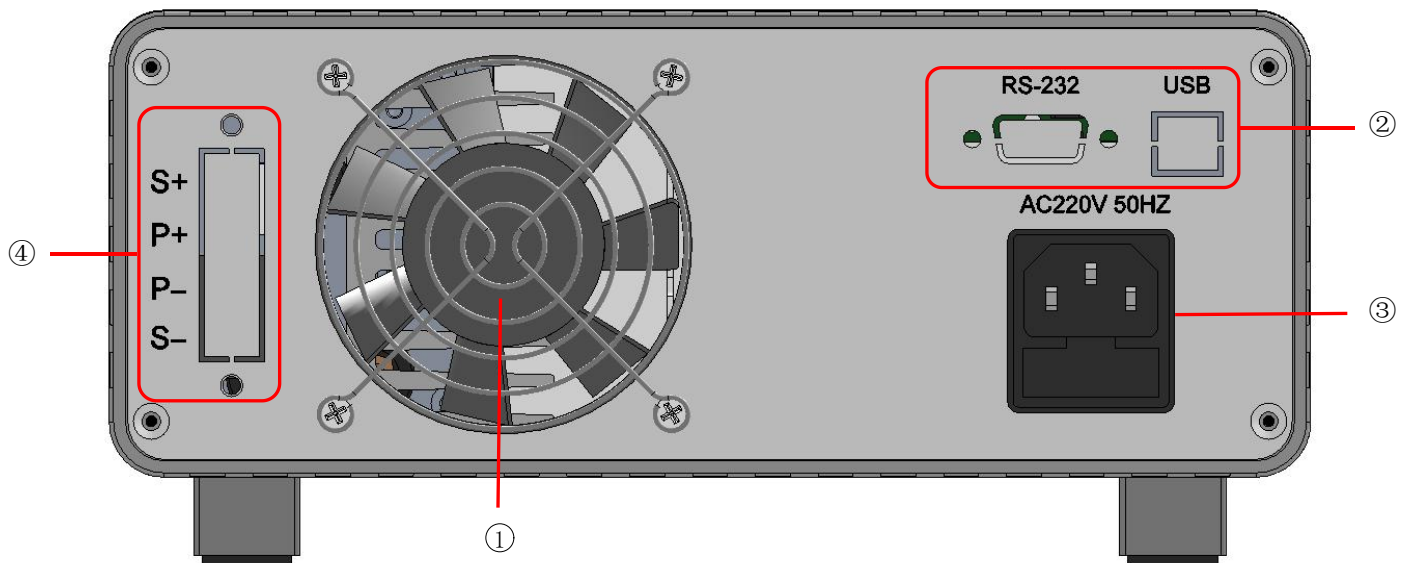


Figure 2: Rear panel

- ① Heat dissipation window
- ② Communication interface
- ③ Power input socket
- ④ Spare terminal

3.2 Basic operation

3.2.1 Booting

Connect the ASD-906B analog battery to the mains and press the power switch. At this time, the LCD displays the boot LOGO. After about 10 seconds of preheating, the analog battery is initialized and enters the standby interface. At this time, it has been successfully turned on. After turning on the LCD Display, as shown in Figure 3 and Figure 4.



Figure 3: LCD display interface when starting up



Figure 4: LCD display interface after booting

3.2.2 Voltage setting

After the analog battery is turned on and enter the standby interface, first confirm whether the set voltage is the required voltage, if not, long press the digital area below the voltage setting value to call up the voltage setting page, as shown in Figure 5.



Figure 5: Voltage setting display

3.2.3 Output enable

After the simulated battery is turned on and enter the standby interface, confirm that the output voltage is correct and press the "⏻" button in the function key area to enable the set voltage output. At this time, the "⏻" button will light up, as shown in Figure 6.

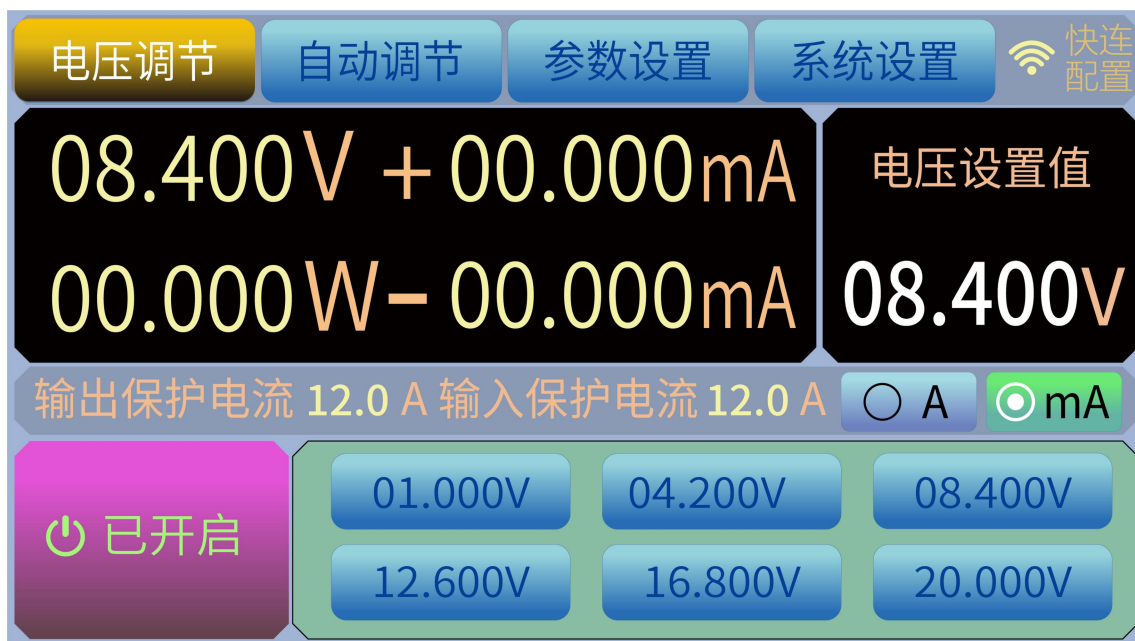


Figure 6: Display interface after output is enabled

Tip: When the output is enabled, directly click the 6 voltage settings below to quickly switch the required voltage. Long press the voltage value area to set the preset voltage. You can also enter the real-time voltage adjustment function by directly pressing the adjustment knob. Press the adjustment knob again to select the adjustment position. After 5 seconds of no adjustment operation, it will automatically exit the real-time adjustment function and save the adjusted display interface.

3.2.4 Automatic adjustment settings

Press the "automatic adjustment" in the function setting area to enter the interval voltage automatic adjustment setting interface, as shown in Figure 7, click the position where the initial voltage value needs to be set, and manually enter the required initial voltage; click on the required Set the end voltage value position, manually enter the required cut-off voltage; click the position where the step voltage value needs to be set, and manually enter the required step voltage; click the position where the step time value needs to be set, and manually enter the step time ; Check the stop condition box to enable the conditional stop function; click the position where the current value needs to be set to be less than and manually enter the current value; click the position where the delayed start time needs to be set, and manually enter the delayed start time; turn on the output switch Icon, click the execute button, the analog battery will automatically adjust the voltage output according to the conditions set by the automatic adjustment.



Figure 7: Automatic adjustment display interface

3.2. 5 Charge and discharge over current protection value setting

Press the "protection setting" in the function setting area to enter the "charge/discharge over current protection" value setting interface, as shown in Figure 8 , click the location of the over current protection value to be set, and manually enter the required over current Just confirm the protection value. The charge/discharge current protection time on this page can be set to simulate battery delay protection. After all the parameters are set, click the OK button to return to the test interface.

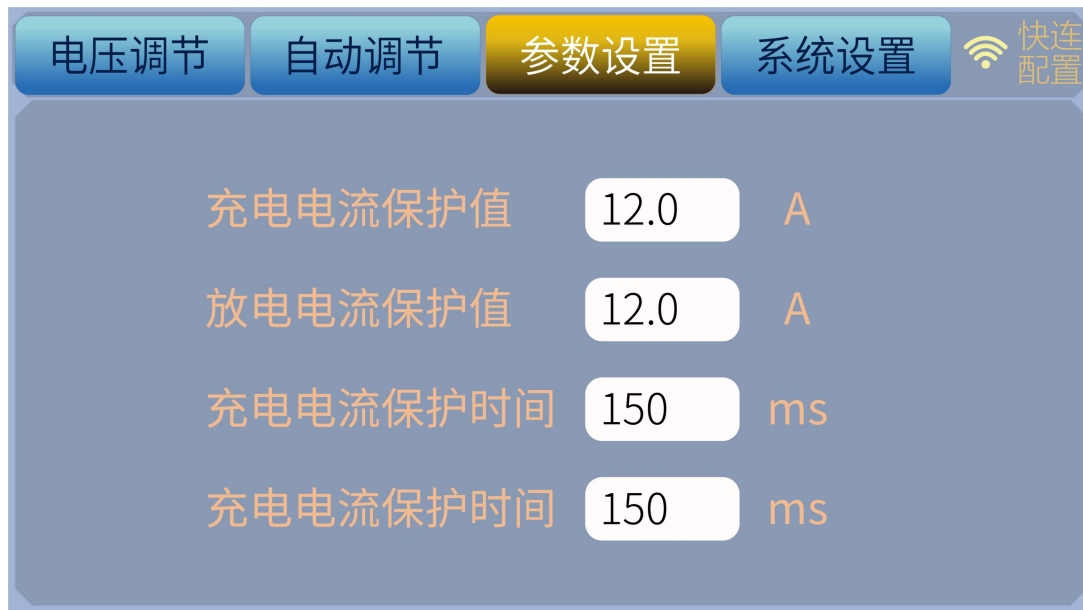


Figure 8 : Charge and discharge over current protection display interface

3.2. 6 System Settings

Press the function setting region "system setting" the mode that is set to enter the system interface, as shown in Fig 9 shown, Type display machine model, Temp show the internal temperature,

Panel software ver
Panel hardware ver
Modul software ver
Modul hardware ver



Figure 9 : Function setting display interface



Figure 10 : Standard Alligator Clip Wiring

4.Environmental requirements

Item	Parameter	Specifications
Work environment	Temperature	0 ° C-55 ° C (Please use derating if the ambient temperature exceeds 40°C)
	Humidity	Max: 85%
	Dust	Pollution degree 2
	Altitude	Below 2000 meters
Storage environment	Temperature	-30°C~70°C
	Humidity	Max: 90%

Warranty Agreement

1. This product provides a one-year warranty from the date of purchase (according to the date of issuance of the bill).
2. The following conditions are not covered by the warranty:
 - ①After purchase, the machine is damaged due to improper transportation, use or storage (liquid, damp, squeezed by external force, falling, etc.);
 - ②Repair and alteration not approved by our company;
 - ③Machine damage caused by natural disasters (such as lightning, earthquake, fire, flood, etc.) and secondary disasters;
 - ④Failure and damage caused by factors other than the machine;
 - ⑤Incomplete procedures for warranty card and purchase receipt;
 - ⑥Product accessories are not covered by the warranty.
3. When the product fails or is damaged, please fill in the contents of the "Product Warranty Card" correctly and in detail.
4. The "Product Warranty Card" will not be reissued under normal circumstances, please keep this card properly.
5. After the warranty period expires, in order to provide you with more lasting and perfect services, we will provide paid maintenance services.
6. For the collection of maintenance fees, please refer to the latest version of the "Maintenance Price List" of our company.
7. If you have any questions during the service, please contact our agent or our company in time.
8. The right to interpret this agreement belongs to Shenzhen ASunDar Electronics Co., Ltd.



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